

MORRASALE, Russian FederationWMO: **230320**Lat: **69.714N**Lon: **66.814E**Elev: **26**StdP: **101.01**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-38.2	-35.9	-40.7	0.1	-38.3	-38.4	0.1	-35.8	19.4	-16.4	16.9	-17.2	5.4	60	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	19.9	16.2	17.3	14.2	14.7	12.4	16.5	19.5	14.4	16.8	12.5	14.5	5.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.6	18.5	13.1	9.4	16.2	11.3	8.4	13.6	46.4	19.5	40.7	16.8	35.7	14.5	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.2	12.5	DB	-40.9	22.9	3.5	3.4	-43.4	25.3	-45.5	27.3	-47.5	29.2	-50.1	31.7	(4)
(5)			WB	-39.9	17.7	4.1	2.2	-42.9	19.3	-45.3	20.5	-47.6	21.8	-50.6	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.6	-19.7	-20.7	-16.4	-11.7	-4.8	3.3	8.4	7.6	4.3	-2.4	-11.9	-15.8	(6)	
	DBStd	12.24	8.53	9.41	8.41	6.96	4.65	4.30	5.32	3.35	3.20	5.38	7.39	9.10	(7)	
	HDD10.0	6121	922	858	819	652	457	207	97	92	174	386	657	799	(8)	
	HDD18.3	9093	1181	1092	1077	902	716	450	311	334	422	644	907	1058	(9)	
	CDD10.0	73	0	0	0	0	0	7	48	16	2	0	0	0	(10)	
	CDD18.3	4	0	0	0	0	0	0	4	0	0	0	0	0	(11)	
	CDH23.3	15	0	0	0	0	0	0	15	0	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
	WSAvg	6.3	6.6	6.4	6.4	6.4	6.3	5.8	5.3	5.6	6.0	6.9	7.0	6.9	(14)	
(15) Precipitation	PrecAvg	257	14	12	11	12	15	25	30	42	43	26	16	16	(15)	
	PrecMax	436	46	30	32	35	45	58	56	97	92	55	38	42	(16)	
	PrecMin	139	1	2	2	2	3	2	6	3	6	11	4	4	(17)	
	PrecStd	80	10	7	8	8	11	13	16	24	22	12	9	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	-1.3	-0.2	2.0	7.8	19.7	25.4	19.4	13.4	6.6	1.1	0.0	(19)	
		MCWB	-1.4	-1.7	-0.6	0.8	6.2	14.6	18.1	16.0	11.7	5.8	0.6	-0.4	(20)	
	2%	DB	-3.0	-3.2	-1.3	0.7	4.0	15.9	21.9	16.5	11.5	5.6	-0.1	-1.1	(21)	
		MCWB	-3.2	-3.6	-1.6	0.2	2.9	12.0	17.7	14.1	10.4	4.9	-0.6	-1.6	(22)	
	5%	DB	-5.4	-5.0	-2.5	0.1	2.1	13.0	19.3	14.3	10.0	4.7	-1.0	-2.3	(23)	
		MCWB	-5.8	-5.3	-2.8	-0.3	1.4	10.2	15.8	12.7	9.1	4.1	-1.5	-2.7	(24)	
	10%	DB	-7.7	-7.6	-4.3	-1.2	0.9	10.1	16.8	12.4	8.7	3.8	-2.2	-3.6	(25)	
		MCWB	-8.1	-7.9	-4.6	-1.7	0.4	8.2	13.9	11.1	7.9	3.3	-2.6	-3.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-1.6	-0.5	1.0	6.0	14.7	19.1	16.6	12.2	6.1	0.7	-0.3	(27)	
		MCDB	-1.2	-1.4	-0.3	1.4	7.1	18.6	23.3	18.9	13.2	6.5	0.8	-0.2	(28)	
	2%	WB	-3.3	-3.6	-1.5	0.3	2.9	12.4	17.7	14.4	10.6	5.1	-0.4	-1.5	(29)	
		MCDB	-3.1	-3.3	-1.3	0.7	3.7	15.6	21.5	16.1	11.3	5.5	-0.2	-1.2	(30)	
	5%	WB	-5.8	-5.3	-2.8	-0.2	1.4	10.3	16.1	12.8	9.2	4.2	-1.4	-2.6	(31)	
		MCDB	-5.5	-5.0	-2.5	0.1	2.0	12.8	19.2	14.2	9.9	4.6	-1.1	-2.3	(32)	
	10%	WB	-8.1	-7.9	-4.6	-1.7	0.5	8.2	14.2	11.3	7.9	3.3	-2.6	-3.9	(33)	
		MCDB	-7.7	-7.6	-4.3	-1.2	0.9	10.0	16.6	12.3	8.6	3.8	-2.2	-3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	6.6	6.9	6.9	4.6	5.0	6.0	4.2	3.4	3.8	6.3	7.0	(35)	
		MCDBR	7.6	7.5	6.2	6.0	4.9	10.4	9.6	8.2	5.0	2.8	4.7	6.4	(36)	
	5% WB	MCWBR	7.6	7.3	6.2	5.8	4.2	7.3	5.9	5.9	3.9	2.7	4.9	6.4	(37)	
		MCWBR	7.7	7.3	6.1	5.8	4.5	10.1	9.8	7.5	4.5	2.7	4.7	6.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.205	0.239	0.290	0.316	0.330	0.342	0.370	0.355	0.313	0.252	N/A	N/A	(40)	
		taud	1.853	1.960	2.022	2.039	2.099	2.288	2.372	2.431	2.458	2.177	N/A	N/A	(41)	
	Edn at Noon		11	466	692	803	842	848	806	776	710	468	0	0	(42)	
		Edh at Noon	3	45	87	117	126	108	97	82	59	35	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.01	0.34	1.92	4.10	5.87	5.99	5.07	3.10	1.55	0.42	0.00	0.00	(44)	
	RadStd		0.00	0.06	0.16	0.22	0.30	0.48	0.69	0.35	0.13	0.06	0.00	0.00	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+1.72	+2.78	+3.36	+1.64	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page